

K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS),SAMAYAPURAM-TRICHY

DEPARTMENT OF AI&DS

CGB1122-DATA STRUCTURES



CALENDAR MANAGEMENT SYSTEM

PRESENTED BY:

SAGAYA PRINCY J

8115U25D090

SUPERVISOR:

Mrs.G.ASHA

ASSISTANT PROFESSOR/AIDS

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INTRODUCTION

- **Calendar Management System** is a software application used to organize and manage events, meetings, appointments, and reminders.
- It helps users keep track of their daily, weekly, and monthly schedules in an efficient way.
- The system allows users to add, update, delete, and view events stored in the calendar.
- It reduces the difficulty of managing schedules manually and helps avoid missing important tasks or meetings.
- In this project, data structures are used to store and manage calendar events efficiently.
- Structures such as **arrays, linked lists, trees, or hash tables** can be used

OBJECTIVE

- To create a system for managing calendar events and schedules efficiently.
- To allow users to add new events, meetings, and appointments.
- To provide the facility to edit or update existing event details.
- To enable users to delete events that are no longer needed.
- To store event information in an organized manner using data structures.
- To allow quick searching of events based on date, time, or event name.
- To display daily, weekly, and monthly schedules clearly.

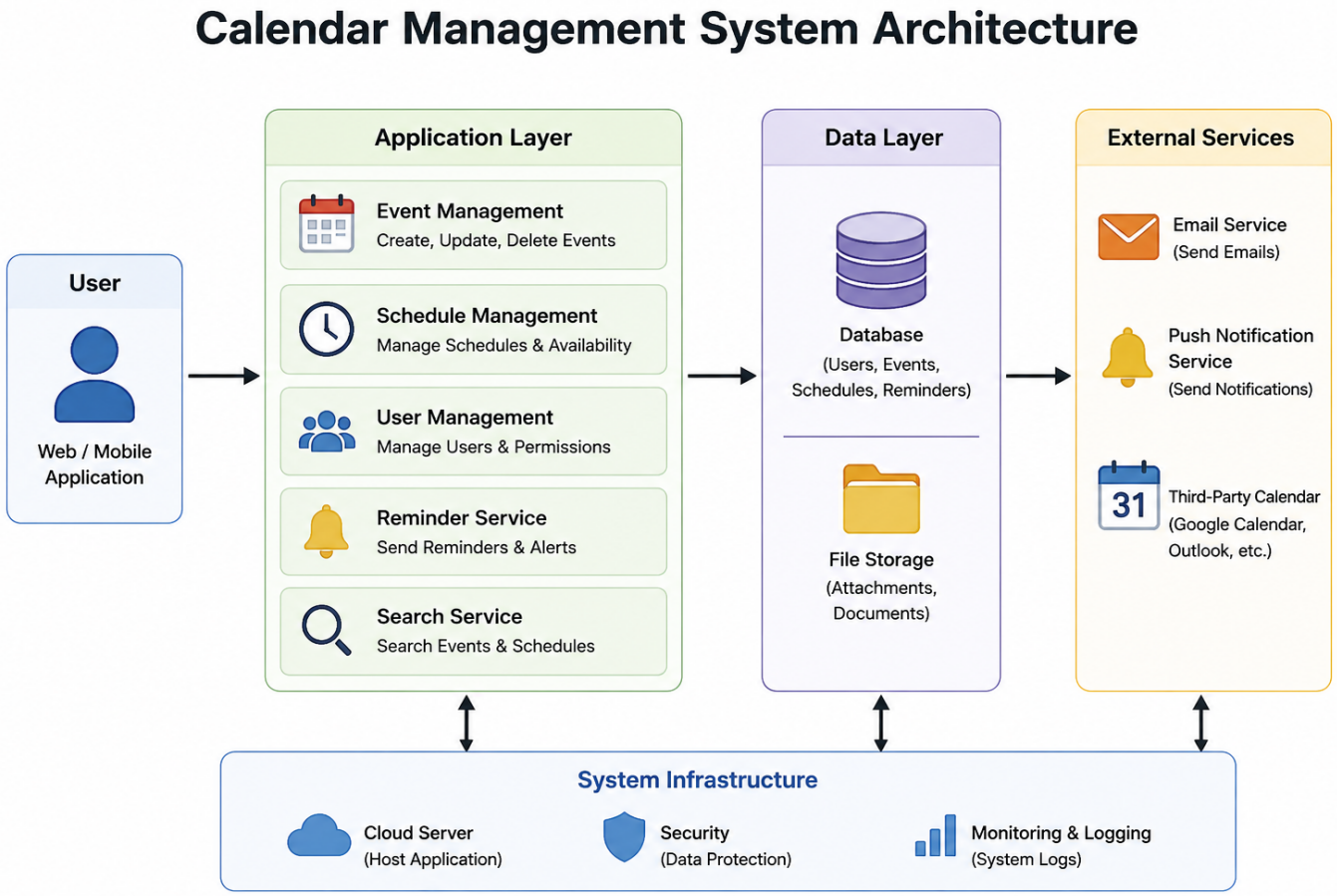
PROBLEM STATEMENT

- Many people find it difficult to manage their daily schedules, appointments, and important events manually.
- Traditional methods such as paper calendars and notebooks can lead to missed appointments, lost information, and scheduling conflicts.
- As the number of events increases, searching, updating, and organizing schedules becomes time-consuming.
- There is a need for a system that can efficiently store, manage, and retrieve calendar events.
- Therefore, the Calendar Management System is developed to provide an organized, user-friendly, and efficient solution for managing schedules and events.

DATA STRUCTURES USED

- Queue follows the FIFO (First In First Out) principle.
- Events are added using the Enqueue operation.
- Events can be removed using the Dequeue operation.
- The first event entered is processed first.
- Queue helps manage calendar events in chronological order.
- It is useful for scheduling and organizing upcoming events efficiently.
- Queue uses Front and Rear pointers to manage data.

ARCHITECTURE DIAGRAM



MODULES USED

- USER MANAGEMENT MODULE
- EVENT MANAGEMENT MODULE
- SCHEDULE & REMINDER MODULE
- CALENDAR STORAGE MODULE

MODULE 1 – USER MANAGEMENT MODULE

Functionality

Handles user registration ,login and profile management.

Process

- Register new users
- Login authentication
- Manage user profiles

Security

- Validates user credentials
- Prevents unauthorized access

MODULE 2 – EVENT MANAGEMENT MODULE

Functionality

Manages creation, modification and deletion of calendar events.

Process

- Create events
- Update events details
- Delete events

Security

- Prevents duplicate entries
- Validates event information

MODULE 3 – SCHEDULE & REMINDER MODULE

Functionality

Organizes schedules and sends reminders for upcoming events.

Process

- Generate notification
- Sct reminders

Security

- Ensures reminder accuracy
- Protects schedule information

MODULE 4 – CALENDAR STORAGE MODULE

Functionality

Stores and retrieves calendar events efficiently using data structure.

Process

- Stores event records
- Retrieve events details
- Maintain event history

Security

- Prevents stack overflow and underflow
- Controls memory usage efficiently

OUTPUT SCREENSHOT

```c
#include <stdio.h>
#include <string.h>

#define MAX 5

struct Event {
    int date;
    char title[50];
};

struct Event queue[MAX];
int front = -1, rear = -1;

void enqueue() {
    if (rear == MAX - 1) {
        printf("Queue is Full\n");
        return;
    }

    if (front == -1)
        front = 0;

    rear++;

    printf("Enter Date: ");
    scanf("%d", &queue[rear].date);

    printf("Enter Event Title: ");
    scanf("%s", queue[rear].title);
}

void display() {
    int i;

    if (front == -1) {
        printf("No Events Available\n");
        return;
    }

    printf("\nCalendar Events:\n");
    for (i = front; i <= rear; i++) {
        printf("Date: %d\tEvent: %s\n",
               queue[i].date,
               queue[i].title);
    }
}

int main() {
    int n, i;

    printf("Enter Number of Events: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        enqueue();
    }

    display();

    return 0;
```

Output

```
Enter Number of Events: 3
Enter Date: 5
Enter Event Title: meeting
Enter Date: 12
Enter Event Title: seminar
Enter Date: 20
Enter Event Title: Exam

Calendar Events:
Date: 5 Event: meeting
Date: 12    Event: seminar
Date: 20    Event: Exam


=== Code Execution Successful ===
```

FUTURE SCOPE

This project can be improved in many ways:

- Add AI-based automatic scheduling.
- Access calendar from any device using cloud storage.
- Add voice command support.
- Improve reminder and notification features.
- Support team and group event management.
- Enhance security and data backup.

CONCLUSION

- The project demonstrates the practical use of data structures for storing and managing event data effectively. By using appropriate data structures, the system can perform operations such as insertion, deletion, and searching efficiently.
- Overall, the Calendar Management System improves time management, reduces scheduling conflicts, and provides an organized way to keep track of important activities. This project enhances understanding of data structures and their real-world applications in software development.

REFERENCES

- Reema Thareja, **"Data Structures Using C"**, Oxford University Press.
- Seymour Lipschutz, **"Data Structures"**, McGraw-Hill.
- Mark Allen Weiss, **"Data Structures and Algorithm Analysis in C"**, Pearson Education.
- Horowitz, Sahni, and Anderson-Freed, **"Fundamentals of Data Structures in C"**, Silicon Press.

THANK YOU